

COMP9334 Revision Problems for Week 10A - Solution

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1. The decision variables are

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$$y_i = \begin{cases} 1 & \text{if a controller is to be placed at node } i \\ 0 & \text{otherwise} \end{cases}$$

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$$x_{ij} = \begin{cases} 1 & \text{if the controller at node } i \text{ is to control the switch at node } j \\ 0 & \text{otherwise} \end{cases}$$

The optimisation problem is:

$$\min \sum_{i=1}^n y_i$$

subject to

$$\sum_{i=1}^n x_{ij} = 1 \quad \forall j = 1, \dots, n \quad (1)$$

$$x_{ij} d_{ij} \leq D y_i \quad \forall i, j = 1, \dots, n \quad (2)$$

$$x_{ij} \in \{0, 1\} \quad (3)$$

$$y_i \in \{0, 1\} \quad (4)$$

Equation 1 enforces the constraint that a switch is associated with only one controller. Equation 2 ensures that: (i) If $y_i = 0$, i.e. there is no controller at node i , then $x_{ij} = 0$; (ii) If $y_i = 1$, then any switch that is connected to the controller at node i must have a communication delay of less than D . Equation 2 has combined (i) and (ii) into a set of inequalities. Alternatively, you can use

$$x_{ij} \leq y_i \quad \forall i, j = 1, \dots, n \quad (5)$$

$$x_{ij} d_{ij} \leq D \quad \forall i, j = 1, \dots, n \quad (6)$$

to separately enforce the conditions (i) and (ii).